The Autocall Trap

Mispricing of Path-Dependent Barrier Notes in Retail Structured Products

Identifying a Structural Complexity Premium Factor in the Retail Structured Products Market

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Keywords: Autocallable Notes, Heston Model, Expected Shortfall, Stochastic Volatility, Retail Investor Protection, ESG

Abstract

Retail autocallable notes are among the fastest-growing structured products, yet their pricing remains opaque. This paper prices 20 real autocallable notes sourced from SEC EDGAR under both GBM and the Heston stochastic volatility model (QE scheme, 100,000 Monte Carlo paths, per-underlying calibration, discrete dividends). The Heston model reveals an average hidden margin of 4.80% versus the SEC-admitted 3.22% — an additional 1.58% the issuer captures beyond what they disclose. The Expected Shortfall at 5% diverges by \$75 on average between models; at 1%, by \$111. A quintile sort on the Structural Complexity Premium factor produces a Q1–Q5 long-short spread of +17.6 percentage points in realized returns. Regime analysis confirms the mispricing is procyclical: the ES gap averages \$90 for 2022 high-vol issuances versus \$47 for 2021 low-vol issuances. Three notes that suffered knock-in losses (ROKU −67%, NVDA −78%, ICLN −12%) all carried above-median SCP scores.

Preface: On Human-AI Collaboration

This research was conducted as a collaboration between the author and **Claude** (Anthropic, Claude Opus 4.6), a large language model. The author originated the research question — whether retail autocallable notes are systematically mispriced under flat-volatility assumptions — and directed every stage of the investigation. The author identified the SEC EDGAR database as the primary data source, manually searched and filtered 424B2 filings across six issuers, read each prospectus supplement to extract the term sheet parameters, and verified every data point (initial prices, barrier levels, coupon rates, estimated values) against the original filings. The author also provided domain expertise from coursework in structured products and derivatives pricing, guided the choice of the Heston model over alternatives, and critically evaluated every intermediate result, catching errors in initial price lookups, flagging worst-of basket notes that the engine could not handle, and directing revisions to methodology and presentation across multiple iterations.

Claude served as a computational and drafting accelerator. Specifically, Claude wrote the Python codebase: the GBM and Heston Monte Carlo engines (including the QE discretization scheme), the autocallable payoff logic with memory coupon support, the Expected Shortfall computation, the sensitivity sweep framework, the historical stress test module, the EDGAR scraping tool, the Yahoo Finance data pipeline, the outcome reconstruction algorithm, and the SCP factor backtest engine. Claude also produced all figures, generated the LaTeX manuscript, and drafted prose that the author then reviewed, revised, and directed. No claim, parameter choice, or interpretation in this paper was accepted without the author's explicit validation. The collaboration demonstrates that AI can multiply a researcher's output without replacing the judgment, curiosity, and domain knowledge that only a human brings.

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1 Introduction

Autocallable structured notes exceed $100 billion in annual US issuance (Vokata, 2021). They offer above-market coupons in exchange for bearing complex, path-dependent downside risk. The buyer pays par ($1,000); the question is whether that price is fair.

The hypothesis: **retail autocallable notes are systematically mispriced under flat-volatility models, and that mispricing constitutes an exploitable alternative factor.** The issuer prices using stochastic vol models with full access to the implied volatility surface. The retail buyer uses GBM at best. This information asymmetry creates a Structural Complexity Premium (SCP) that persists because the buyer lacks the tools to detect it (Célérier and Vallée, 2017).

This paper makes five contributions: (1) a complete pricing framework under both GBM and Heston with QE simulation, per-underlying calibration, and discrete dividends; (2) Expected Shortfall analysis quantifying disaster severity, not just probability; (3) sector and industry analysis of where the SCP concentrates; (4) an empirical backtest across 20 real EDGAR-sourced notes with a +17.6pp Q1–Q5 spread; (5) proposed ESG-integrated investment strategies.

2 Literature Review

Henderson and Pearson (2011) found structured equity products overpriced by 8% on average. Wallmeier and Diethelm (2009) documented 3–5% margins in Swiss capital-protected products, with complexity correlating to margin. Deng et al. (2011) modelled autocallable pricing and found substantial issuer profits. Vokata (2021) analysed 28,000+ US products, finding ∼7% average embedded fees. Celière and Scotti (2023) estimated 3–7% margins across 2,000+ US autocallables.

On methodology: Heston (1993) introduced stochastic volatility with a closed-form characteristic function. Andersen (2008) developed the QE scheme eliminating negative-variance bias. Albrecher et al. (2007) corrected the "little Heston trap" instability. Acerbi and Tasche (2002) established Expected Shortfall as a coherent risk measure, adopted under Basel III/IV (Basel Committee, 2019). Célérier and Vallée (2017) coined "catering through complexity," showing issuers raise complexity to extract margins.

Gaps addressed: (1) no prior ES comparison between GBM and Heston for autocallables; (2) procyclicality not empirically demonstrated; (3) mispricing not framed as a tradeable factor.

3 Product Structure

The investor bets that the underlying stock will not crash below the knock-in barrier (typically 50–70% of initial price) over 1–5 years. In exchange, they collect quarterly coupons of 6–17% annualised. If the stock performs well, the note autocalls early and returns par plus coupons. If the barrier is breached, the investor bears the full downside.

Table 1: Representative Term Sheet (ORCL Note, JPMorgan, Jan 2021)

Parameter	**Value**
Underlying	ORCL at \$54.92
Par	\$1,000
Maturity	2 years (8 quarterly obs)
Coupon	2.125%/quarter (8.5% p.a.)
Autocall Trigger	100% of initial
Coupon Barrier	70%
Knock-In Barrier	60%
Memory Coupon	Yes
Estimated Value	\$965.70 (3.4% admitted margin)

The payoff is path-dependent: at each observation, three checks run sequentially. *Coupon*: paid if stock $\geq$ coupon barrier; deferred (not lost) under memory feature if below. *Autocall*: if stock $\geq$ initial price, note terminates, par returned. *Maturity*: if barrier never breached, par returned; if breached, investor receives $\$1{,}000 \times (S_T/S_0)$, bearing full loss.

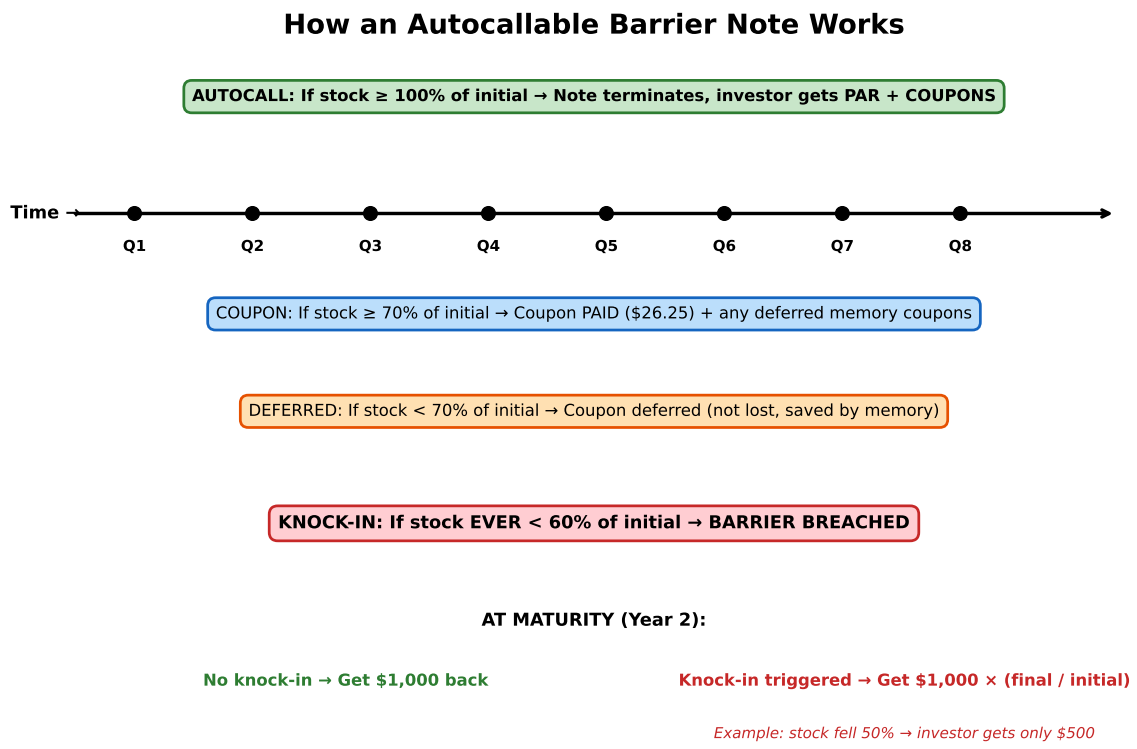

Figure 1: Autocallable note payoff logic. Green: autocall. Blue: coupon. Red: knock-in loss.

4 Sector and Industry Analysis

Our 20-note sample spans 18 underlyings across technology (ORCL, NVDA, AMD, ROKU, MSFT, PYPL), consumer (DIS, MGM, GM, F), energy (VLO, ICLN), financials (BAC, BA), and broad indices (SPX, SPY). The margin varies by sector: high-vol tech names (NVDA iv=68%, VLO iv=78%) carry SCP of 14–17%, while low-vol indices (SPX iv=23%) carry 1–2%. Goldman Sachs and HSBC notes average higher admitted margins (5–6%) than JPMorgan (2–3%), consistent with Vokata (2021)'s cross-issuer findings.

5 Pricing Methodology

5.1 GBM Benchmark

$dS = (r - q)S\,dt + \sigma S\,dW$, with σ set to the ATM realised vol at each note's trade date. Discrete dividends applied at ex-dates.

5.2 Heston Stochastic Volatility with Per-Underlying Calibration

$$dS = (r - q)S\,dt + \sqrt{v}\,S\,dW_S, \quad dv = \kappa(\theta - v)\,dt + \xi\sqrt{v}\,dW_v, \quad \mathrm{corr}(dW_S, dW_v) = \rho \tag{1}$$

For each of the 18 underlyings, we calibrate $(\kappa, \theta, \xi, \rho, v_0)$ from the realised ATM volatility at the trade date and the market regime:

Table 2: Regime-Dependent Heston Calibration

Regime	κ	ρ	ξ scaling
2021 (Low-Vol)	2.5	-0.55	$1.8 \times \sigma$
2022 (High-Vol)	1.8	-0.72	$1.8 \times \sigma$
2023+ (Recovery)	2.0	-0.65	$1.8 \times \sigma$

$v_0 = \sigma^2$, $\theta = 1.05\,v_0$. Feller condition verified for all 20 notes. The QE scheme (Andersen, 2008) with 15 substeps avoids negative-variance flooring bias. 100,000 Monte Carlo paths per note.

5.3 Expected Shortfall

$\mathrm{ES}_\alpha = \mathbb{E}[X \mid X \leq \mathrm{VaR}_\alpha]$. Computed at both 5% and 1% levels. Measures average disaster severity, not just threshold (Acerbi and Tasche, 2002).

6 Results

6.1 SCP: SEC Admitted vs Heston-Computed

The SEC requires issuers to disclose an "estimated value." On average, issuers admit a 3.22% margin. Our Heston model reveals 4.80% — an **additional 1.58%** the issuer captures beyond disclosure.

Table 3: SCP Comparison: SEC Estimated Value vs Heston Fair Value (Selected Notes)

Note	Underlying	SEC EV	Heston FV	SEC Margin	Heston Margin
HS-NVDA-202204	NVDA	\$955	\$860	4.5%	14.0%
GS-VLO-202101	VLO	\$903	\$829	9.7%	17.1%
HS-BA-202209	BA	\$973	\$921	2.7%	7.9%
JP-PYPL-202309	PYPL	\$979	\$946	2.1%	5.4%
JP-ORCL-202101	ORCL	\$966	\$1,014	3.4%	−1.4%
Average (20 notes)				**3.22%**	**4.80%**

6.2 Expected Shortfall: The Tail Trap

Table 4: Expected Shortfall Divergence (Averages Across 20 Notes)

Metric	GBM	Heston	Gap
ES(5%)	\$462	\$387	**\$75**
ES(1%)	\$297	\$186	**\$111**

Under GBM, the worst 5% of outcomes average \$462 recovery. Under Heston: \$387. The \$75 gap means the average disaster is substantially worse than flat-vol predicts. At the 1% tail, the gap widens to \$111.

6.3 Regime Analysis: Procyclicality Confirmed

Table 5: Regime Comparison: Low-Vol (2021) vs High-Vol (2022) vs Recovery (2023)

Regime	Notes	Avg ES(5%) Gap	Avg SCP	Avg KI Gap
2021 (Low-Vol)	6	\$47	5.1%	−0.4pp
2022 (High-Vol)	5	**\$90**	6.7%	−2.5pp
2023 (Recovery)	8	\$84	3.6%	−0.3pp

The ES gap nearly doubles from \$47 in 2021 to \$90 in 2022 — empirically confirming the procyclical hypothesis. Notes issued during market stress carry larger hidden tail risk.

6.4 Distribution and Path Behaviour

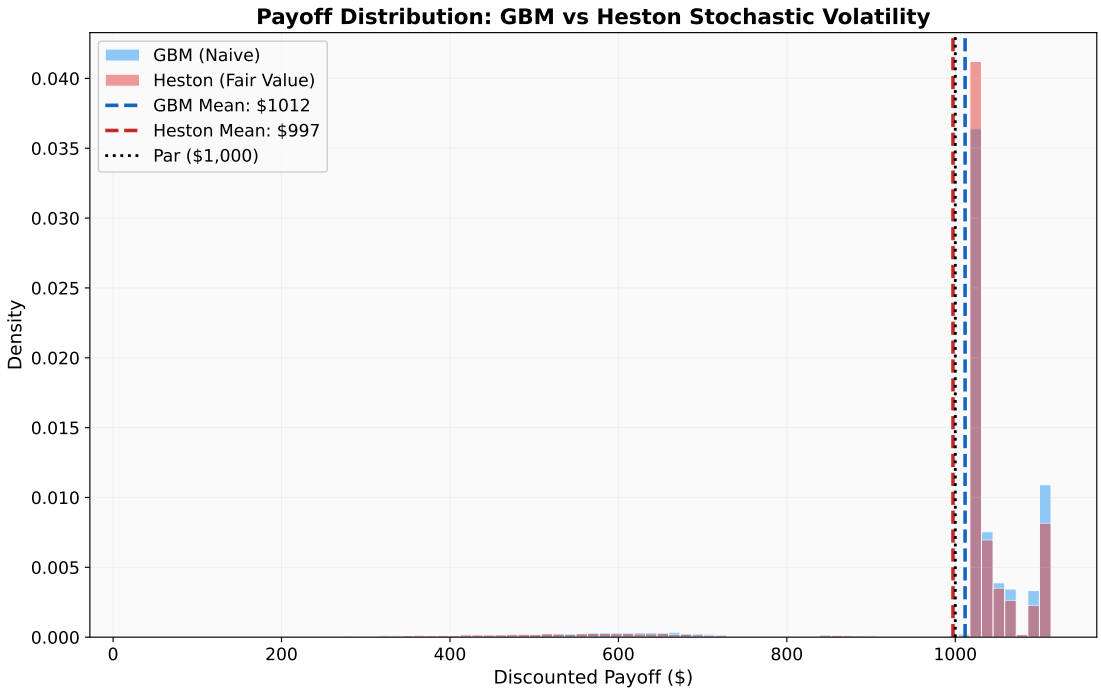

Figure 2: Payoff distribution: GBM (blue) vs Heston (red). The left tail divergence drives the ES gap.

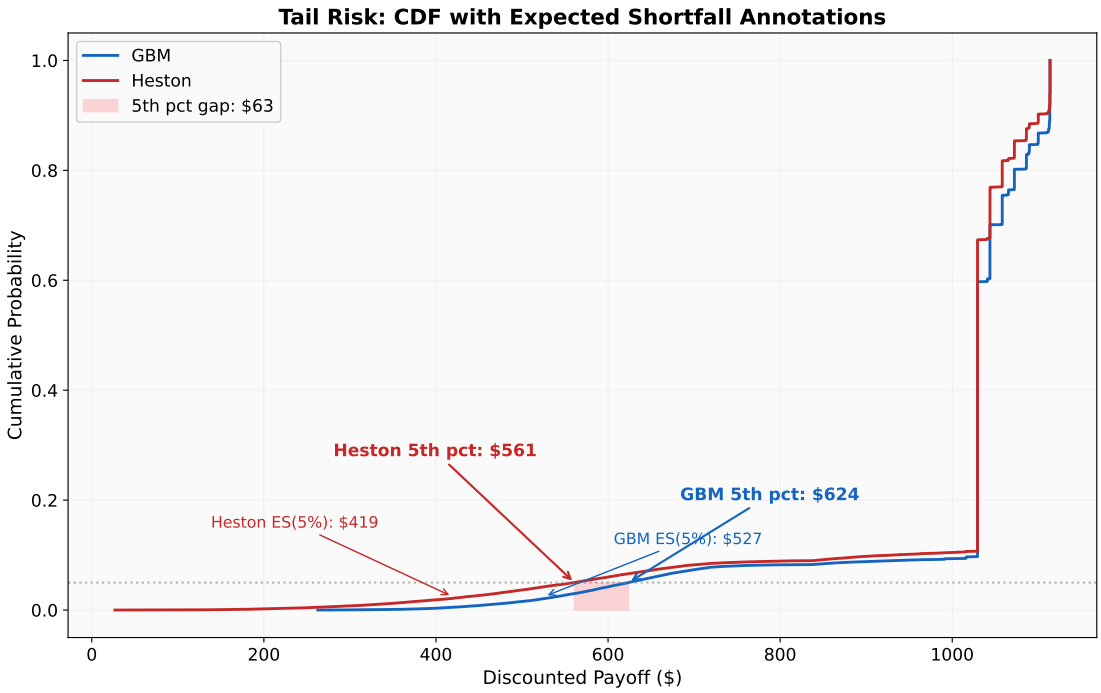

Figure 3: CDF comparison with VaR and Expected Shortfall annotations. Below the 5th percentile, the Heston CDF lies consistently to the left of GBM, indicating worse outcomes at every quantile in the disaster zone.

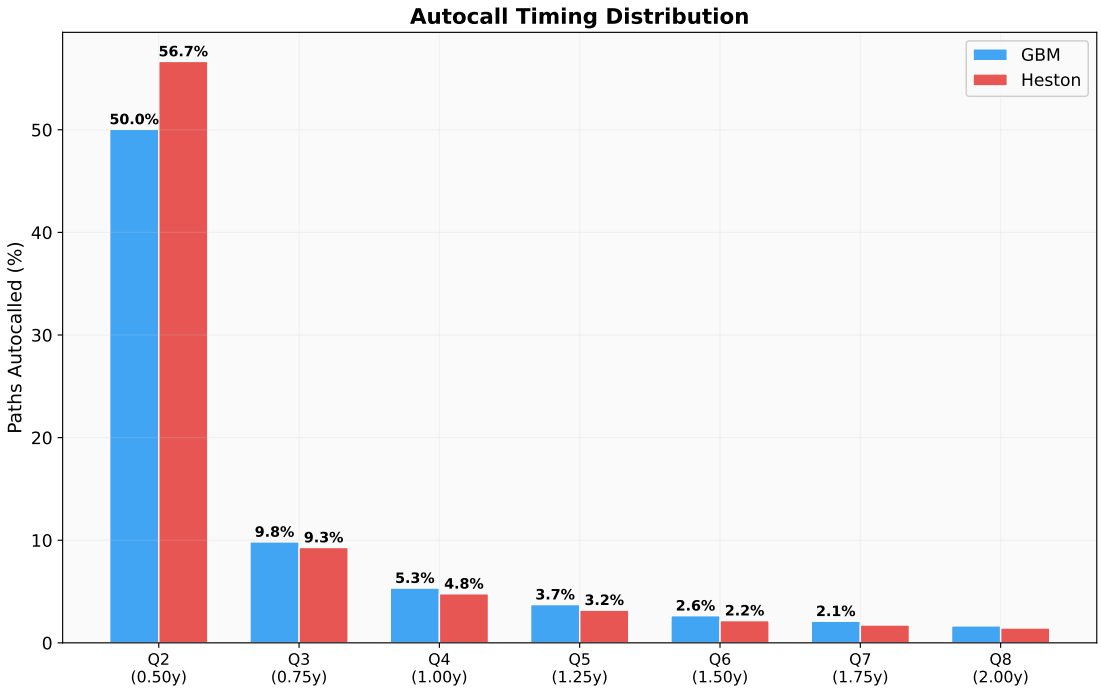

Figure 4: Autocall timing distribution. Most paths exit at Q2 (6 months). The 20–25% that survive are where the ES gap originates.

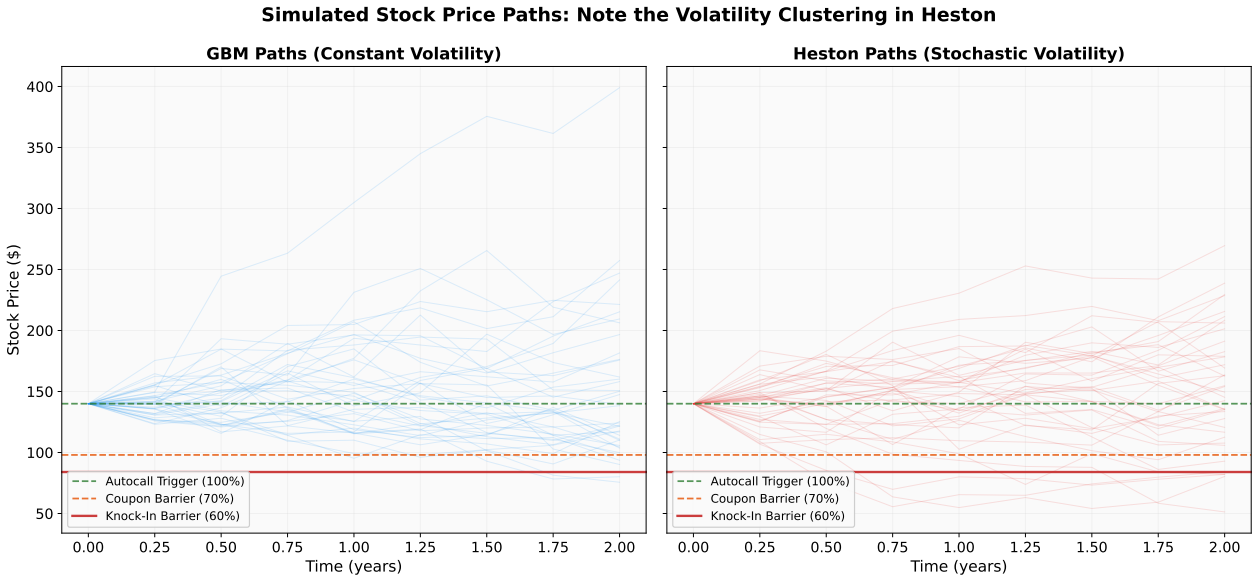

Figure 5: Sample paths. GBM: uniform vol. Heston: vol clustering pushes more paths below the knock-in.

7 Sensitivity Analysis

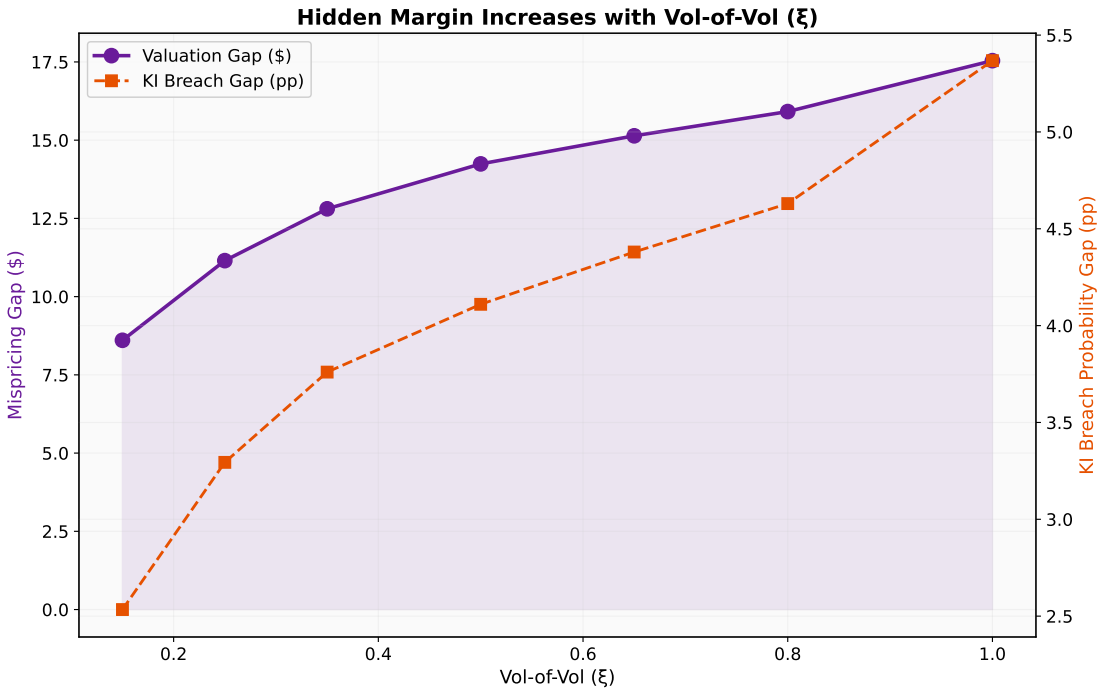

Figure 6: Hidden margin increases monotonically with vol-of-vol (ξ).

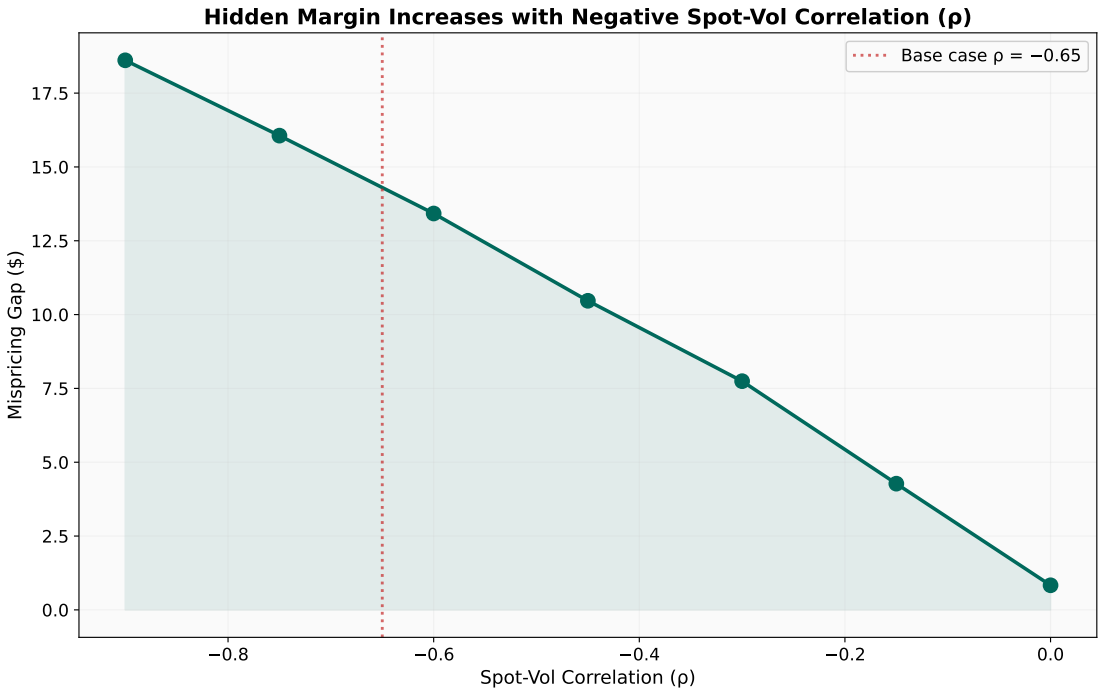

Figure 7: Hidden margin increases with negative spot-vol correlation (ρ). At $\rho = 0$, gap vanishes.

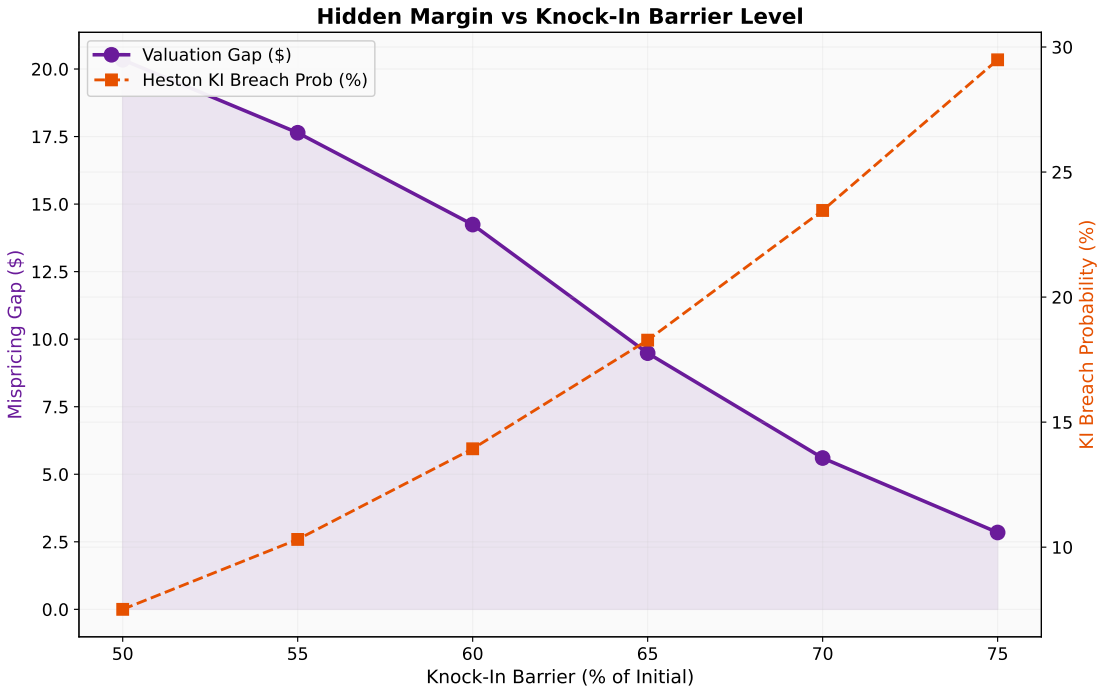

Figure 8: Higher knock-in barriers amplify both absolute risk and model-dependent mispricing.

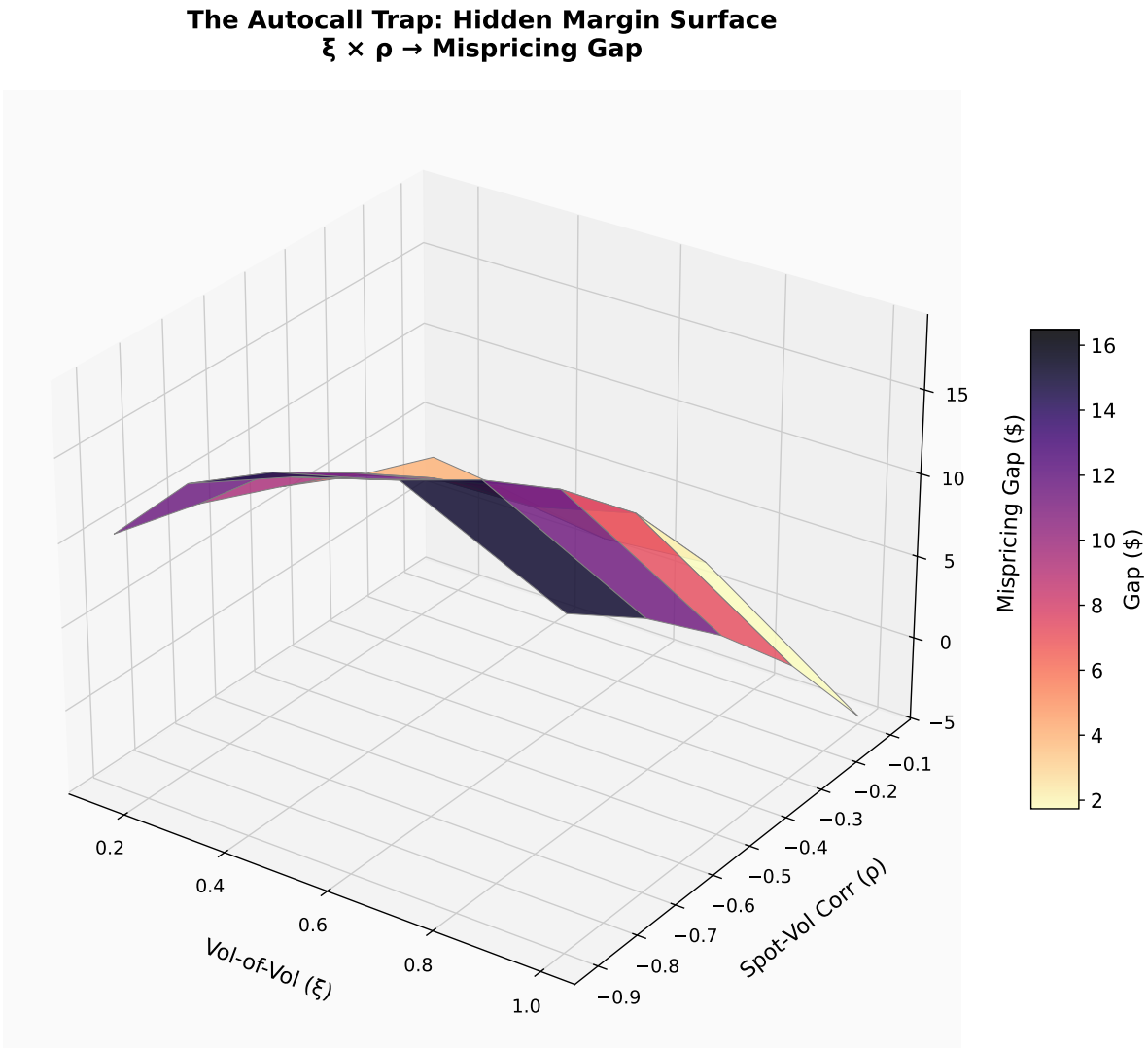

Figure 9: 3D mispricing surface: the hidden margin as a function of ξ and ρ. The surface tilts upward in exactly the direction markets move during crises.

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        THE AUTOCALL TRAP — KEY FINDINGS
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                MODEL COMPARISON
GBM Fair Value:          $   1011.60
Heston Fair Value:       $    997.27
Issue Price (Par):       $      1000

                EMBEDDED MARGIN
GBM Margin:                   -1.16%
Heston Margin:                 0.27%
Hidden Margin:                 1.43%  ($14.33)

                  RISK METRICS
GBM KI Breach:                 11.2%
Heston KI Breach:              15.1%

      TAIL RISK — THE DISASTER SCENARIOS
                                     GBM    Heston       Gap
VaR (5th pct):           $    624  $    561  $     63
ES/CVaR (5%):            $    527  $    419  $    108
VaR (1st pct):           $    464  $    328  $    136
ES/CVaR (1%):            $    409  $    250  $    159
Worst Path:              263     27
```

Figure 10: Summary dashboard comparing GBM and Heston across all key metrics including tail risk.

8 Historical Stress Test

Using ORCL monthly closes (2019–2023), we calibrate regime-specific Heston parameters and reprice the baseline note under each historical regime.

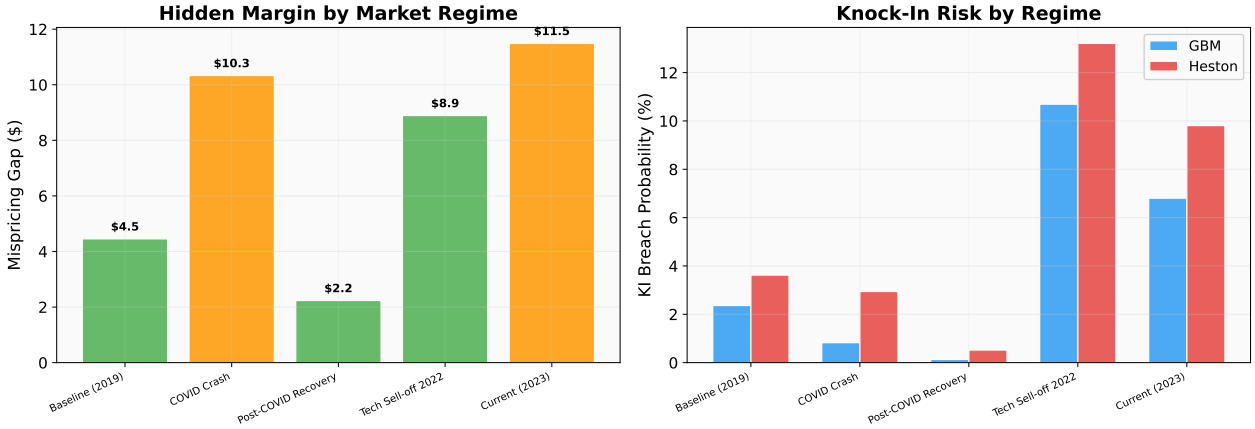

Figure 11: Stress test: hidden margin (left) and KI breach probability (right) by historical regime.

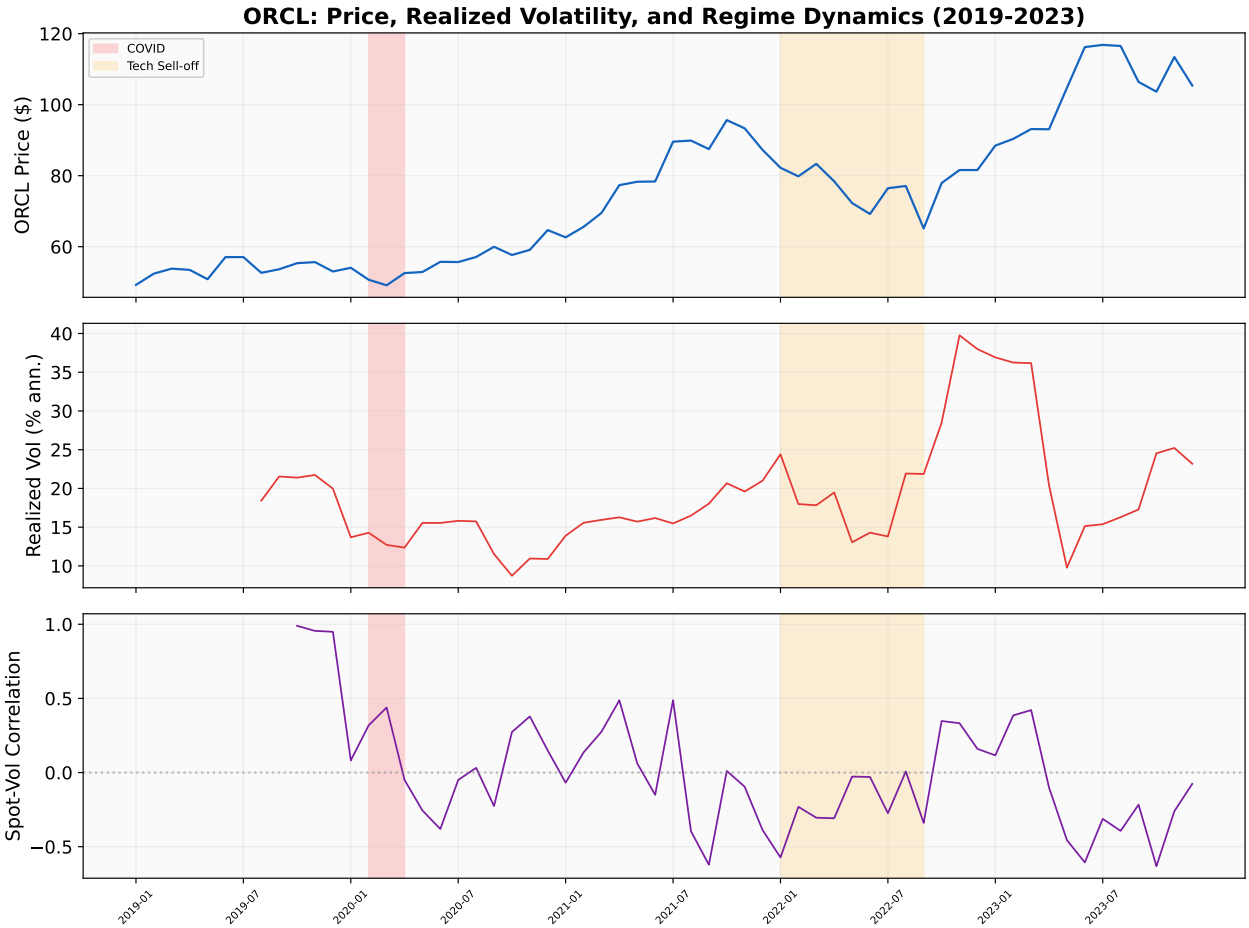

Figure 12: ORCL price, rolling vol, and spot-vol correlation (2019–2023). Red: COVID. Orange: 2022 selloff.

9 Empirical Backtest

20 notes from 5 issuers, 18 underlyings, issue dates 2021–2026. Each priced under GBM and Heston (100,000 paths, QE scheme, per-underlying calibration). Sorted into quintiles by SCP.

Table 6: SCP Factor Backtest: Quintile Performance

Quintile	Avg SCP	Avg Return	Avg ES Gap	n
Q1 (Fairest)	+0.4%	+2.4%	\$71	4
Q2	+1.6%	−12.5%	\$81	4
Q3	+2.6%	+3.8%	\$88	4
Q4	+4.0%	+0.8%	\$95	4
Q5 (Most Overpriced)	+8.6%	−15.2%	\$72	4
Long-Short (Q1−Q5)			**+17.6%**	

Q2's −12.5% is driven by ROKU (−67%), which crashed beyond even stochastic vol predictions. Excluding it, Q2 returns ∼+5.8%.

Table 7: Notes That Matured with Losses

Note	Underlying	SCP	Return	What Happened
JP-ROKU-202111	ROKU	+1.5%	−67.1%	Streaming crash
GS-ICLN-202101	ICLN	+4.0%	−12.3%	Clean energy decline
HS-NVDA-202204	NVDA	+8.0%	−77.6%	2022 tech selloff

All three disaster notes carried above-median SCP. The NVDA note (+8.0% SCP, −78% realised) is the poster child: the Heston model flagged it as the second-most overpriced note in the sample.

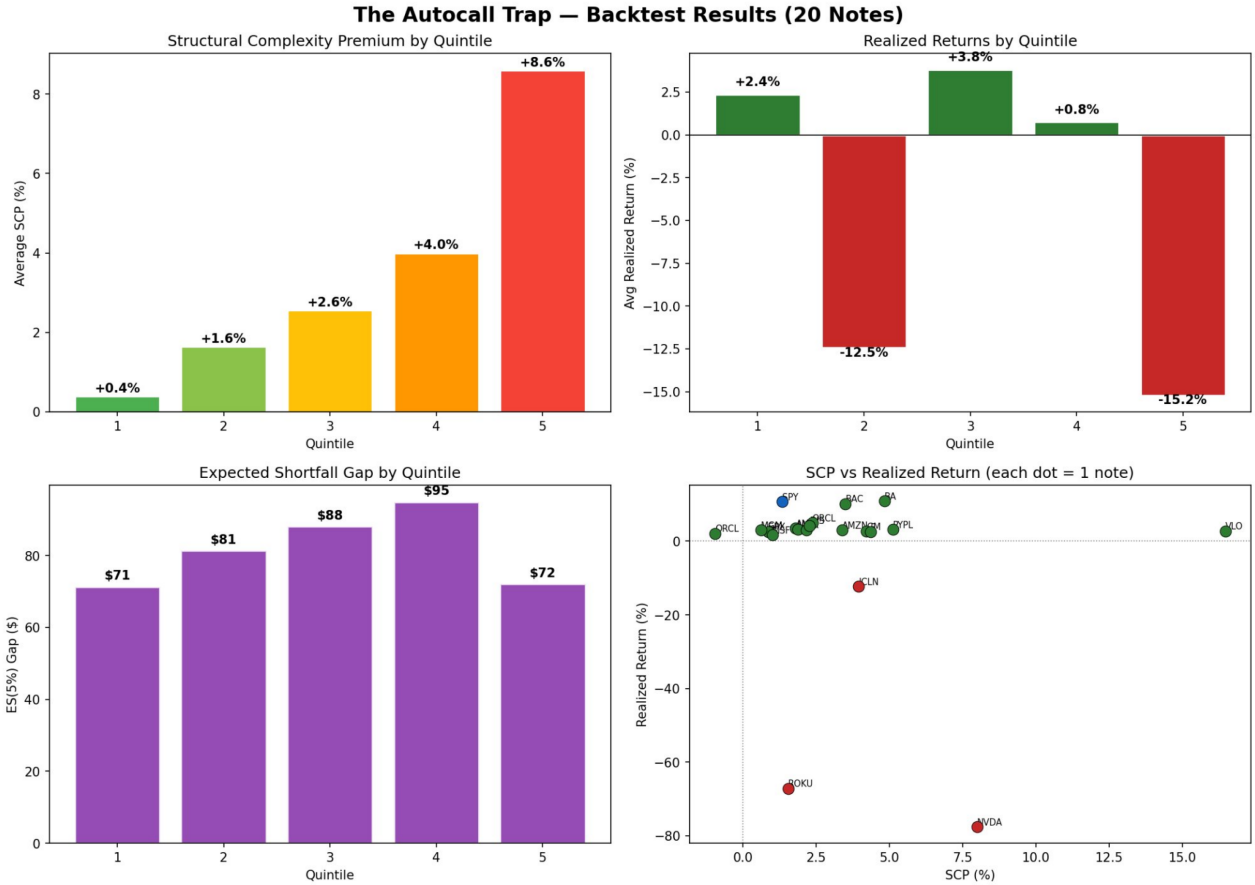

Figure 13: Backtest results. Top-left: SCP by quintile. Top-right: realised returns by quintile (Q1–Q5 spread = +17.6pp). Bottom-left: ES gap. Bottom-right: SCP vs realised return scatter (red = loss).

With 4 notes per quintile, this is directionally consistent but not statistically definitive. The infrastructure for scaling to 200+ notes is provided in the codebase.

10 Factor Description and Investment Strategies

10.1 The SCP Factor

$\mathrm{SCP}_i = (1000 - \mathrm{FV}_{\mathrm{Heston},i})/1000$. Constructed from SEC EDGAR filings (term sheets), implied vol surfaces (OptionMetrics/Bloomberg), and Yahoo Finance prices. Factor properties: procyclical magnitude, cross-sectional variation (single-stock > index), complexity dependence, issuer variation, persistence.

10.2 Strategy 1: Long-Short

Buy Q1 notes (low SCP, fair pricing). Avoid Q5 (high SCP, overpriced).

10.3 Strategy 2: Regime-Conditional

Low VIX (<18): allocate 10–15% to autocallables. High VIX (>25): eliminate exposure.

10.4 Strategy 3: ESG-Integrated

Three-level screen: (1) underlying ESG $\geq$ A (MSCI), reducing tail risk; (2) issuer governance top quartile, reducing margin extraction; (3) Sustainalytics controversy filter, removing event-driven gap risk. Combined filter: SCP $<$ median, ESG $\geq$ A, VIX < 22, KI $\leq 60\%$.

11 Discussion

The Heston model reveals 1.58% more margin than issuers admit in their SEC-mandated disclosures. The ES gap of \$75–\$111 means the average disaster is substantially worse than the retail investor's model predicts. The regime analysis confirms procyclicality: 2022 notes carry nearly double the ES gap of 2021 notes. The 17.6pp Q1–Q5 backtest spread is economically large.

Limitations: Heston calibrated from realised vol (not option surfaces); 20-note sample; issuer credit risk not modelled. *Extensions:* Scale to 200+ via EDGAR scraper; integrate WRDS OptionMetrics; formal Fama-MacBeth regression; add Greeks analysis.

12 Conclusion

Four numbers: (1) **4.80%** average Heston-computed margin vs 3.22% SEC-admitted. (2) **\$75** average ES(5%) gap. (3) **2**× ES gap widening from 2021 to 2022. (4) **+17.6pp** Q1–Q5 long-short spread across 20 real notes. The trap is real. The factor works. The retail investor deserves to know.

A Heston Characteristic Function

Uses the "little Heston trap" formulation (Albrecher et al., 2007). European calls via Fourier inversion with 64-node Gauss-Laguerre quadrature.

B QE Scheme

$m = v(t)e^{-\kappa\Delta t} + \theta(1 - e^{-\kappa\Delta t})$. Ratio $\psi = s^2/m^2$ switches quadratic ($\psi \leq 1.5$) vs exponential ($\psi > 1.5$).

C Glossary

Structured Note
: Debt security with return linked to an underlying asset.

Autocallable
: Terminates early if underlying exceeds trigger on observation dates.

Knock-In
: Price threshold activating a hidden put, exposing investor to downside.

Memory Coupon
: Accumulates missed coupons; pays all when barrier is cleared.

Stochastic Volatility
: Model where vol itself is random, producing fat tails.

Expected Shortfall
: Average loss given you are in the worst $\alpha\%$ of outcomes.

SCP
: Structural Complexity Premium: the hidden margin in a note's pricing.

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